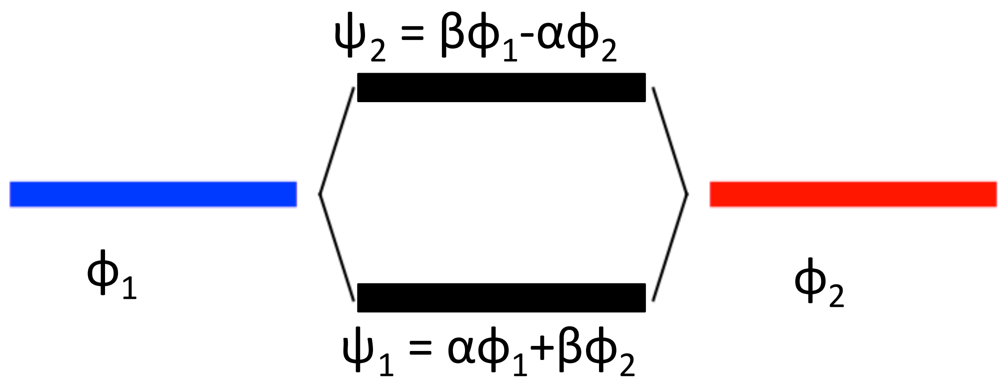
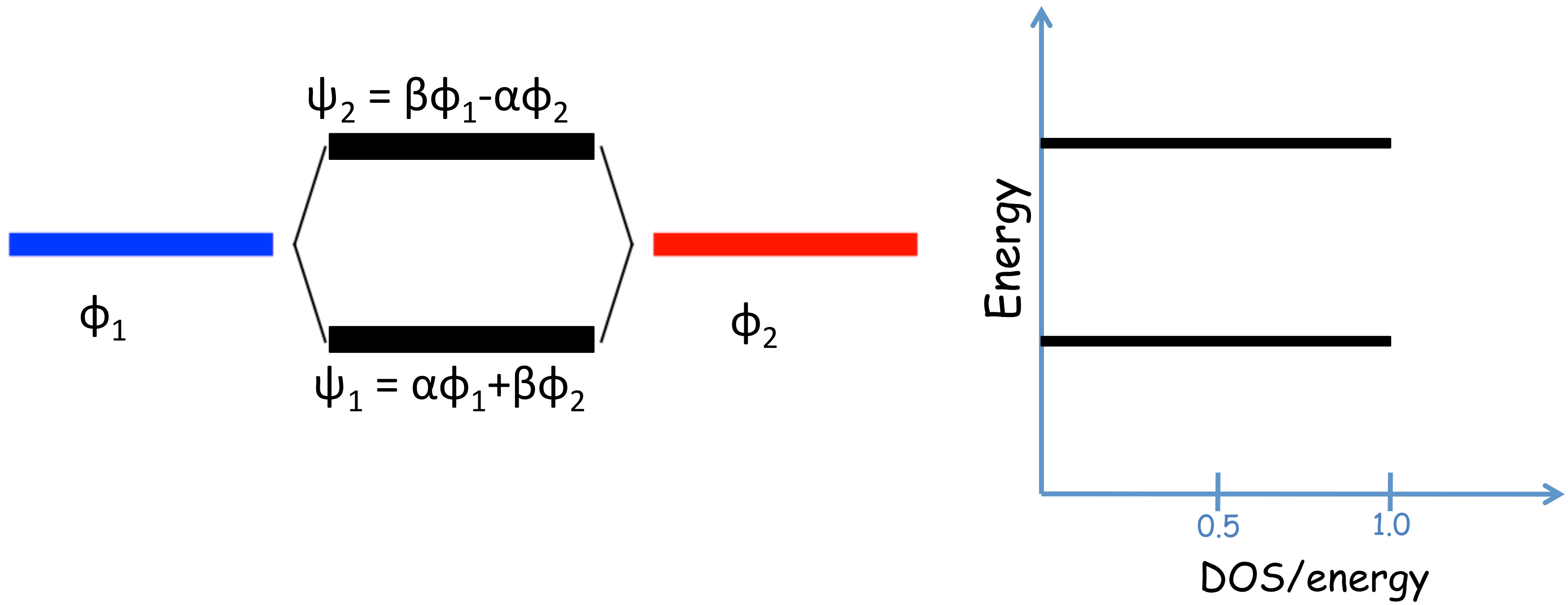
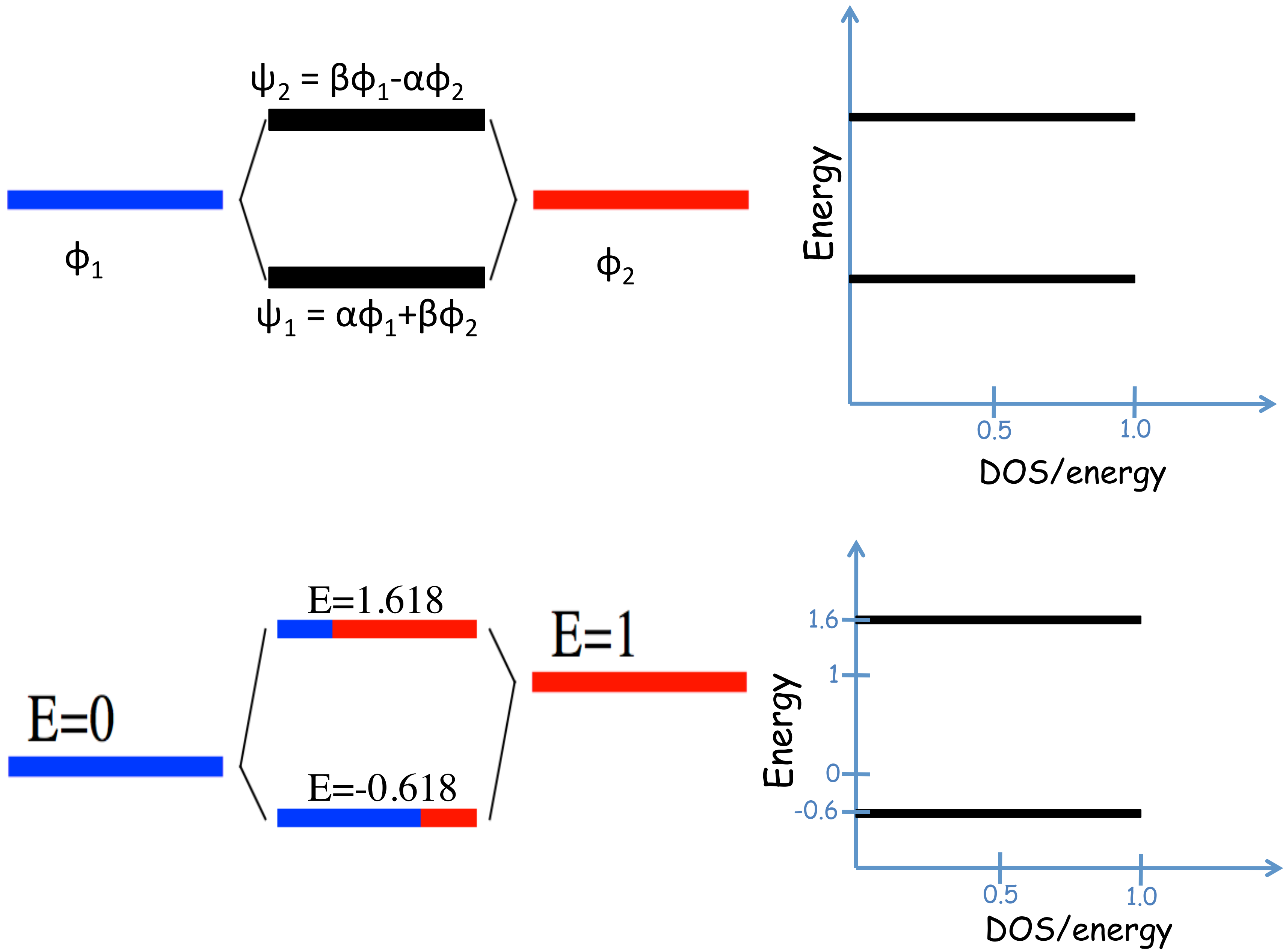
Density of States:

The density of states (DOS) is essentially the number of different states at a particular energy level that electrons are allowed to occupy, i.e. the number of electron states per unit volume per unit energy.


$\psi_2 = \beta\phi_1 - \alpha\phi_2$
ϕ_1
ϕ_2
$\psi_1 = \alpha\phi_1 + \beta\phi_2$


ψ2 = βϕ1-αϕ2
ϕ1
ϕ2
ψ1 = αϕ1+βϕ2
Energy
0.5
1.0
DOS/energy

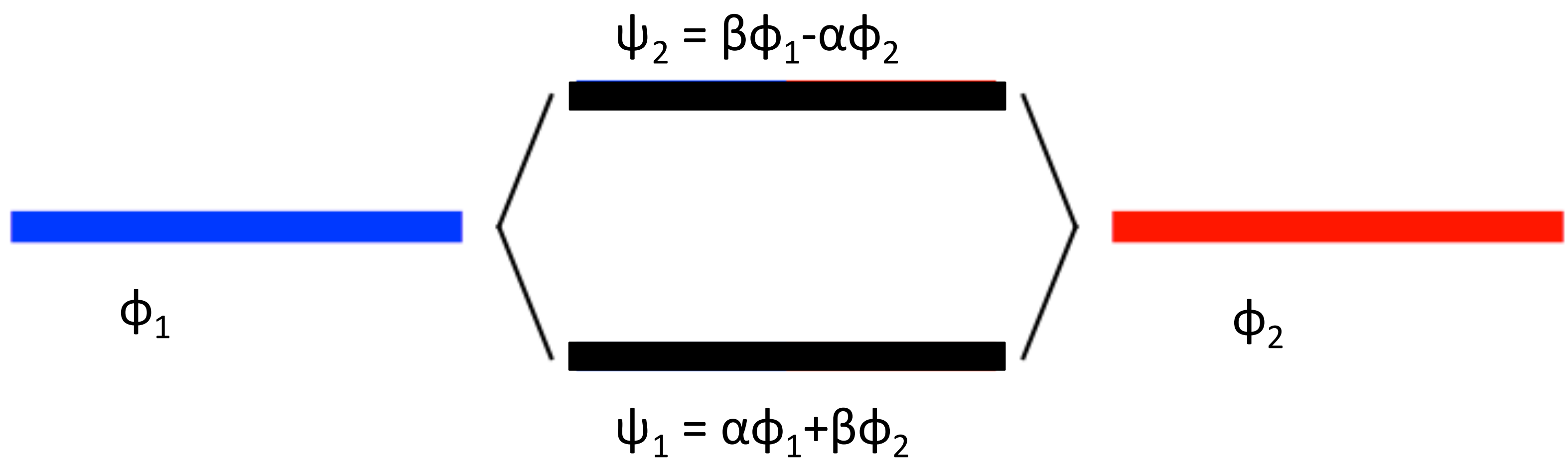

ψ2 = βϕ1-αϕ2
ϕ1
ϕ2
ψ1 = αϕ1+βϕ2
Energy
0.5
1.0
DOS/energy
E=1.618
E=1
E=0
E=-0.618
1.6
1
0
-0.6
Energy
0.5
1.0
DOS/energy

Density of States:

The density of states (DOS) is essentially the number of different states at a particular energy level that electrons are allowed to occupy, i.e. the number of electron states per unit volume per unit energy.

Projected/Partial Density of States:

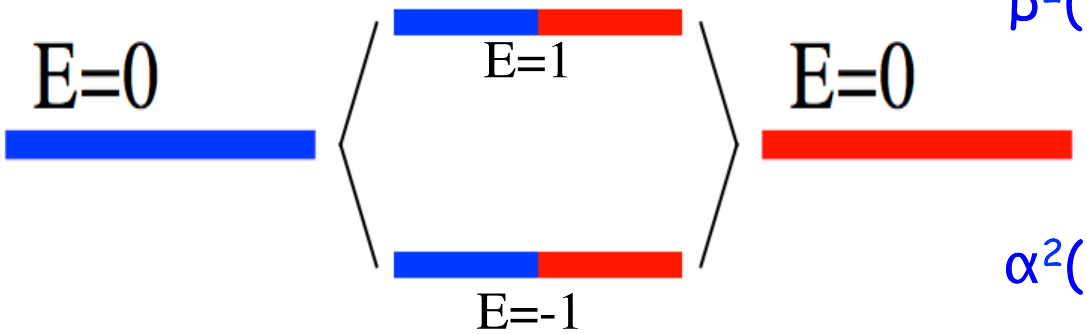
The projected/partial density of states (PDOS) is the relative contribution of a particular atom/orbital to the total DOS



Contribution of ϕ_1 in $\psi_1 = \alpha^2$
Contribution of ϕ_2 in $\psi_2 = \beta^2$

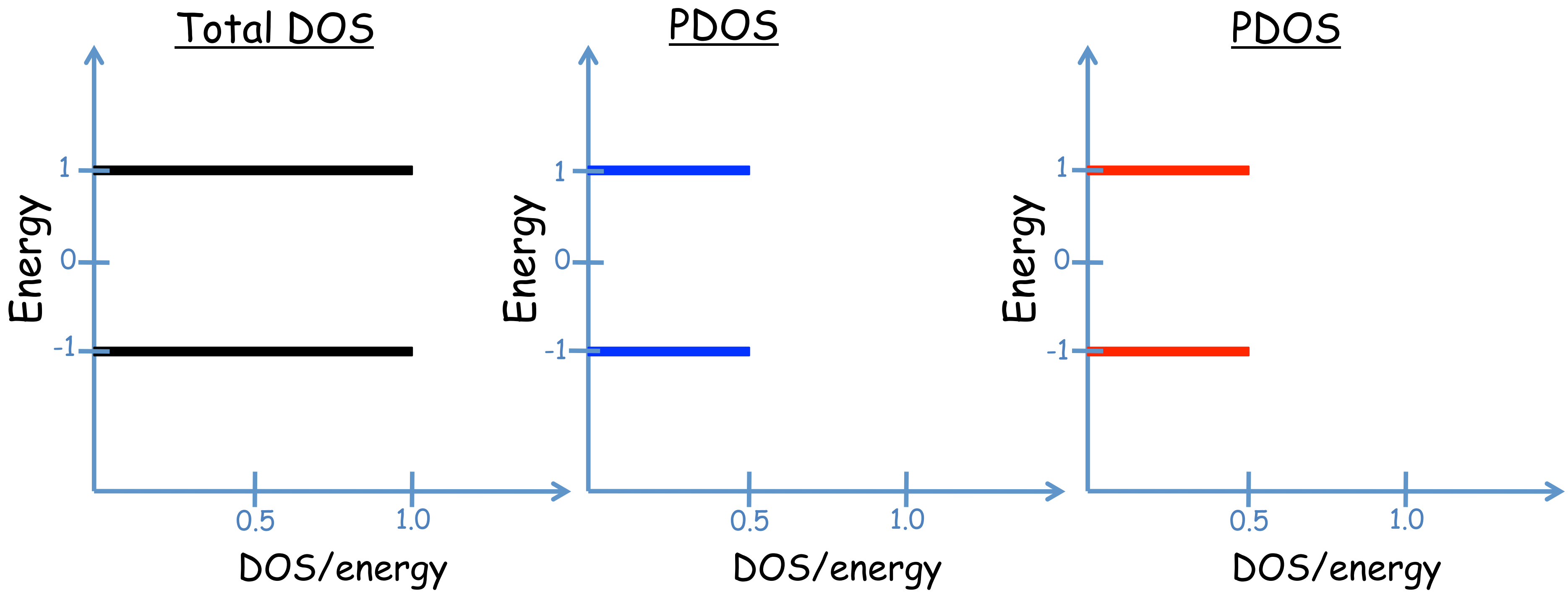
projected density of states

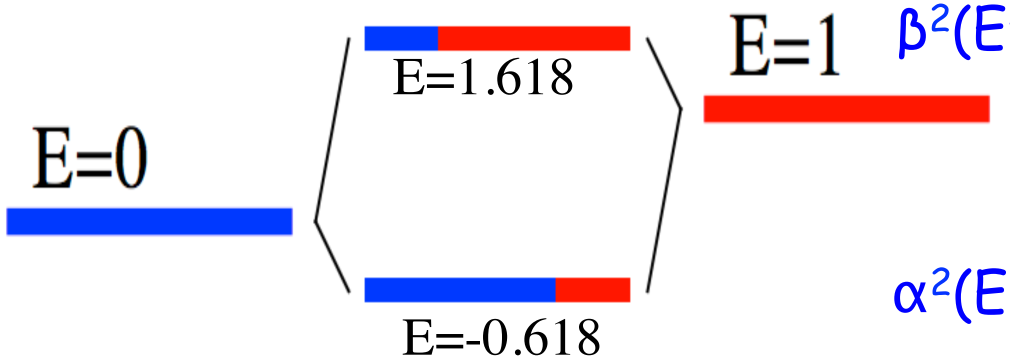
$$\psi_1 = \left\{ \underbrace{\frac{-\Delta - \sqrt{4t^2 + \Delta^2}}{t\sqrt{8 + \frac{2\Delta\left(\Delta + \sqrt{4t^2 + \Delta^2}\right)}{t^2}}}}_{\alpha}, \underbrace{\frac{2}{\sqrt{8 + \frac{2\Delta\left(\Delta + \sqrt{4t^2 + \Delta^2}\right)}{t^2}}}}_{\beta} \right\}$$



$\beta^2(E=0) = 0.5$ $\alpha^2(E=1) = 0.5$

$\alpha^2(E=0) = 0.5$ $\beta^2(E=1) = 0.5$

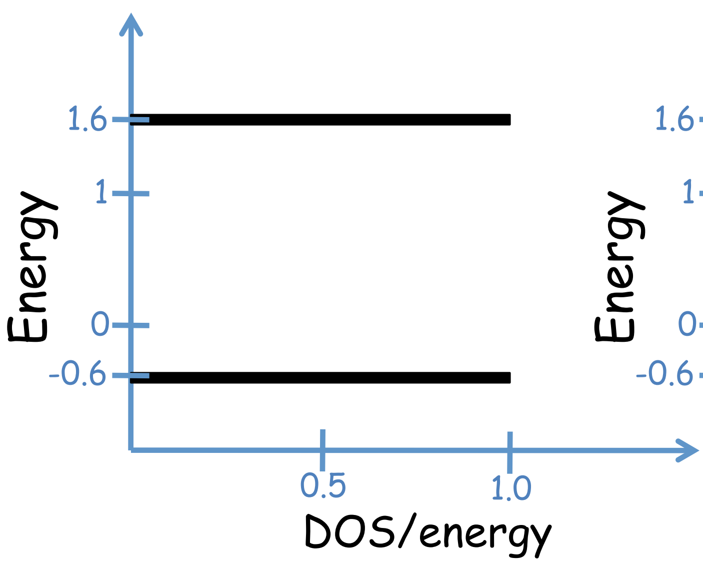




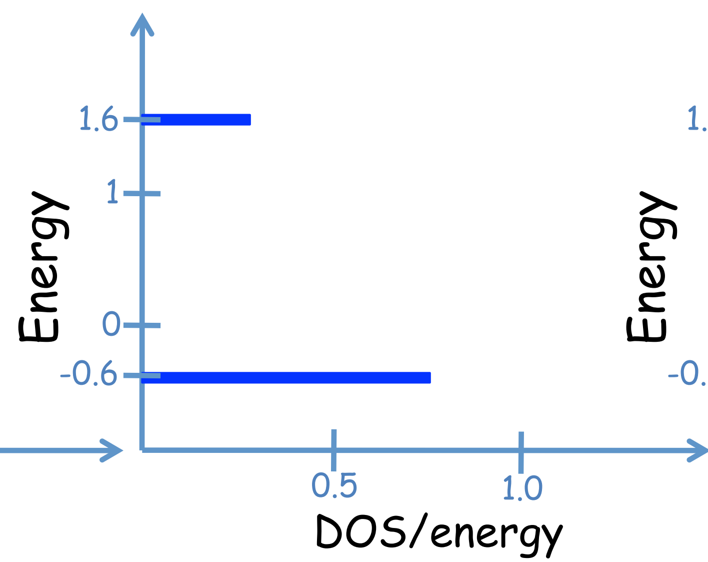
$\beta^2(E=0) = 0.276$ $\alpha^2(E=1) = 0.723$

$\alpha^2(E=0) = 0.723$ $\beta^2(E=1) = 0.276$

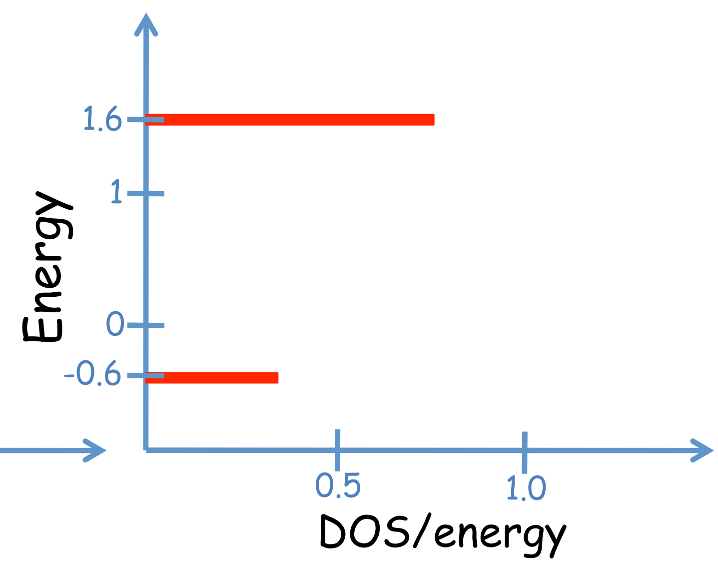
Total DOS



PDOS



PDOS



Density of States:

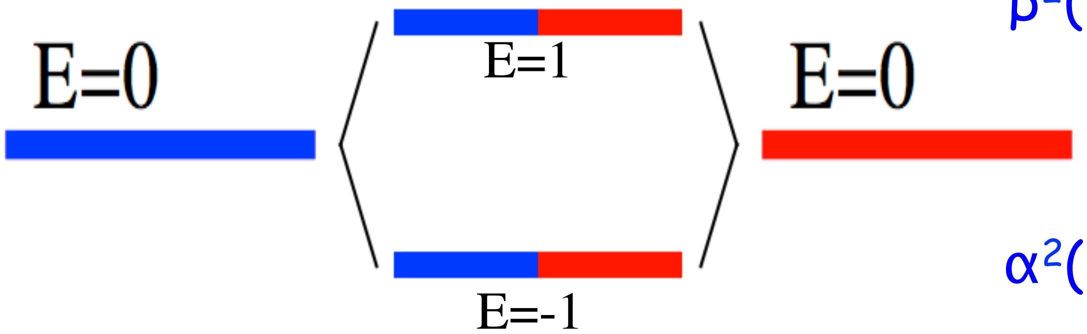
The density of states (DOS) is essentially the number of different states at a particular energy level that electrons are allowed to occupy, i.e. the number of electron states per unit volume per unit energy.

Projected/Partial Density of States:

The projected/partial density of states (PDOS) is the relative contribution of a particular atom/orbital to the total DOS

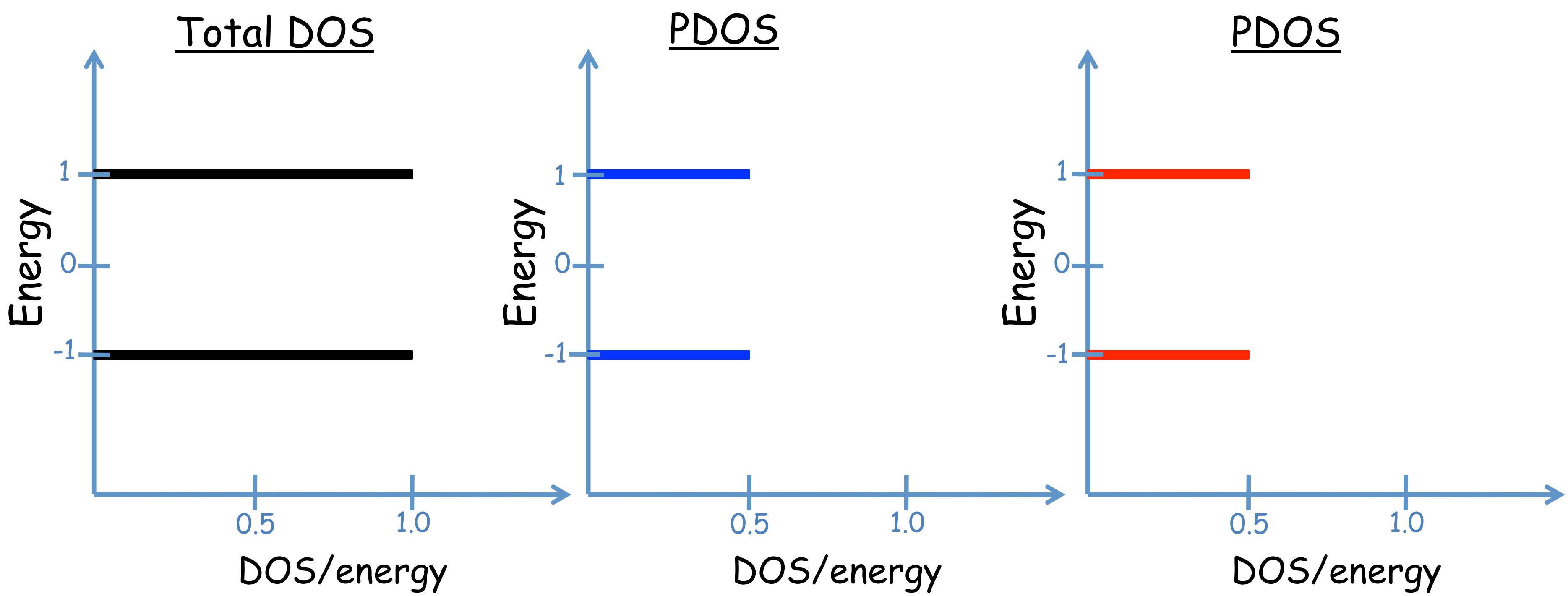
First moment:

The first moment is the center of gravity of PDOS. In other words, first moment is exactly the on-site energy.

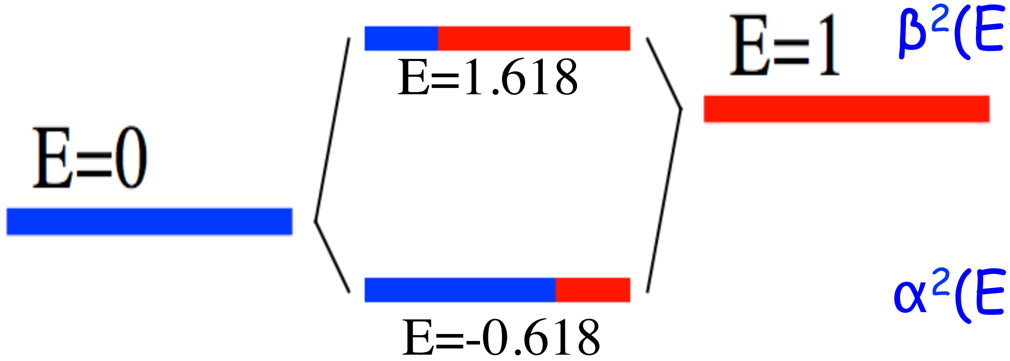


$\beta^2(E=0) = 0.5$ $\alpha^2(E=1) = 0.5$

$\alpha^2(E=0) = 0.5$ $\beta^2(E=1) = 0.5$

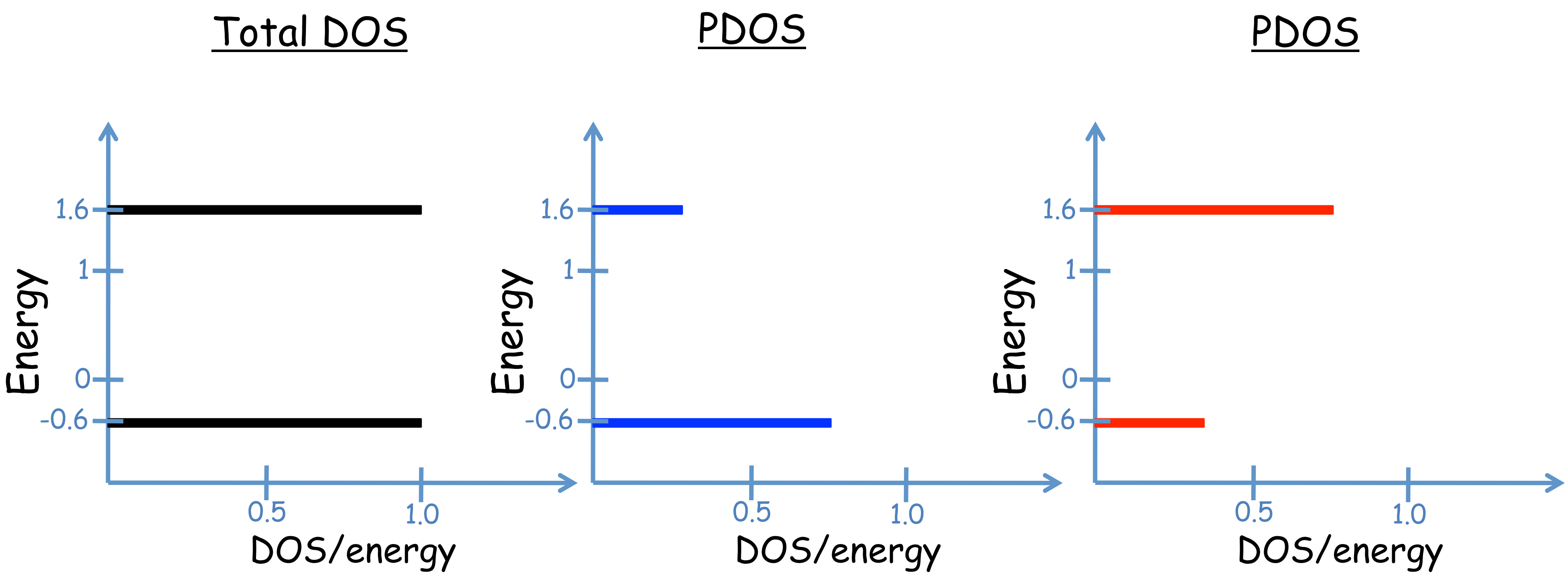


$(-1*0.5) + (1*0.5) = 0$ $(-1*0.5) + (1*0.5) = 0$



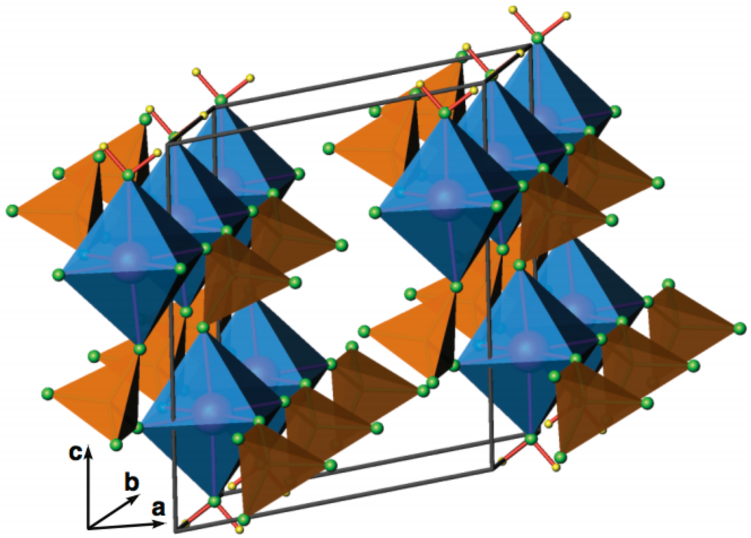
$\beta^2(E=0) = 0.276 \quad \alpha^2(E=1) = 0.723$

$\alpha^2(E=0) = 0.723 \quad \beta^2(E=1) = 0.276$

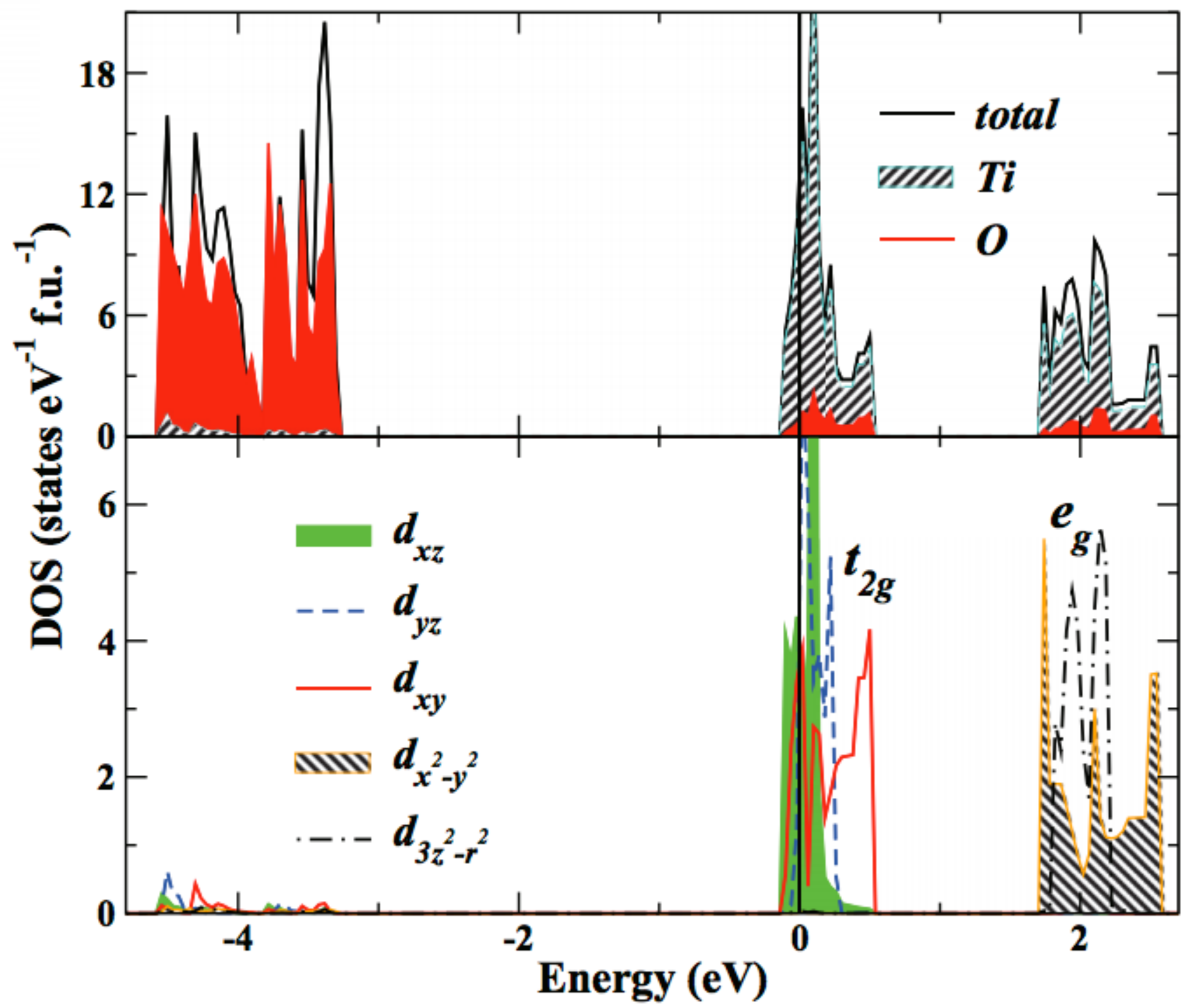


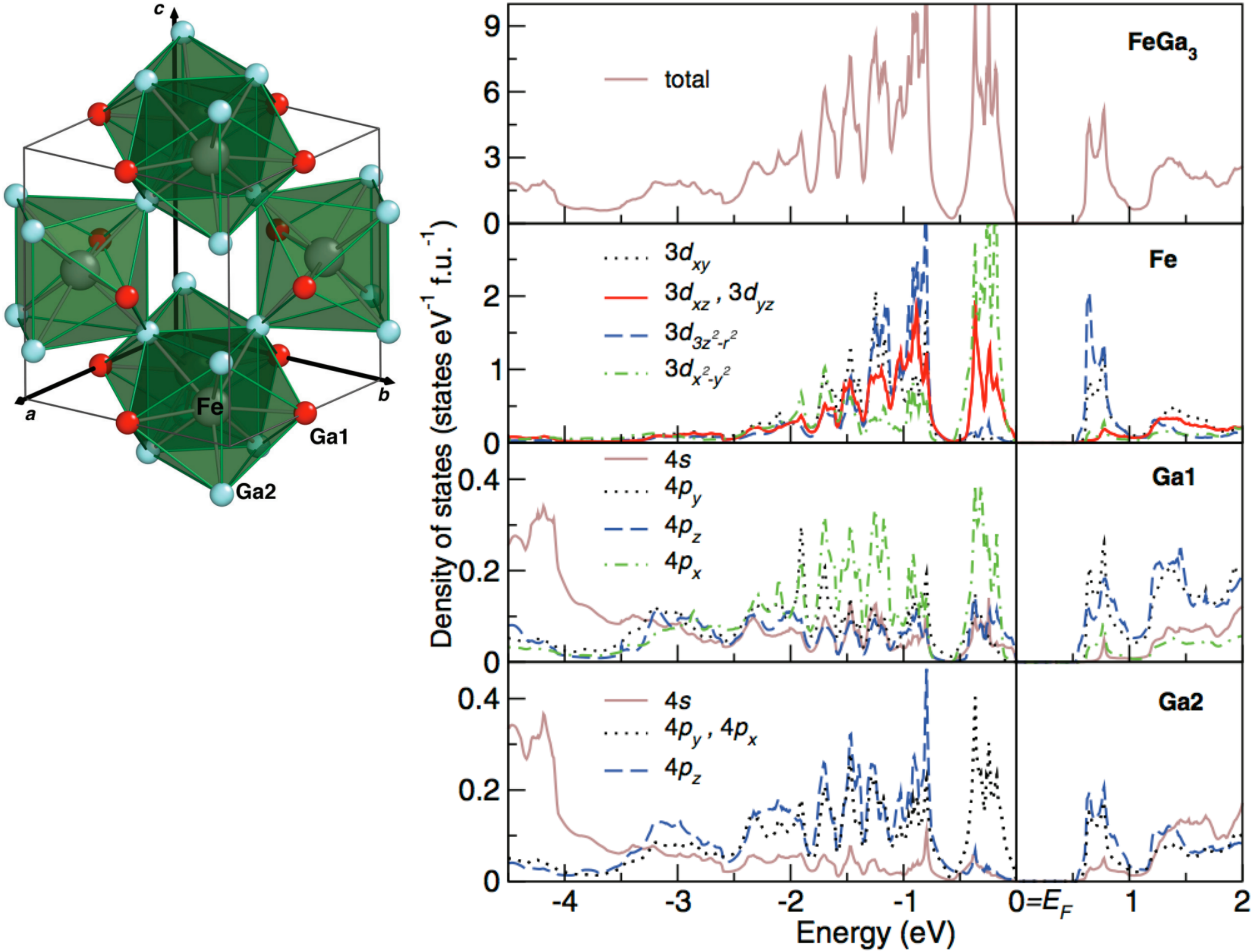
$(-0.618*0.723) + (1.618*0.276) = 0$

$(-0.618*0.276) + (1.618*0.723) = 1$

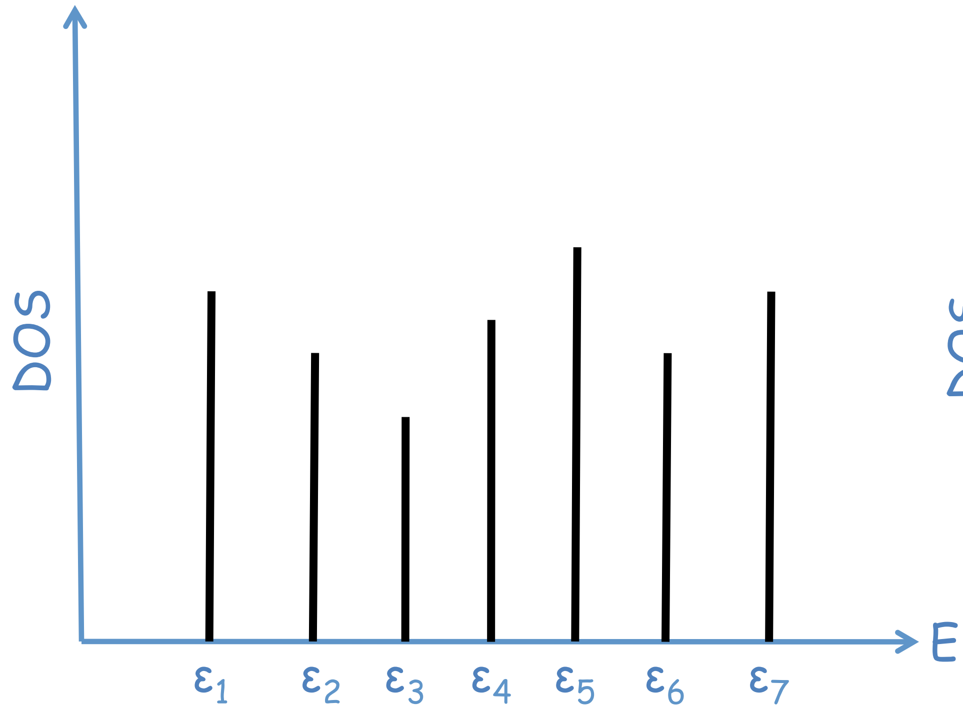

c
b
a

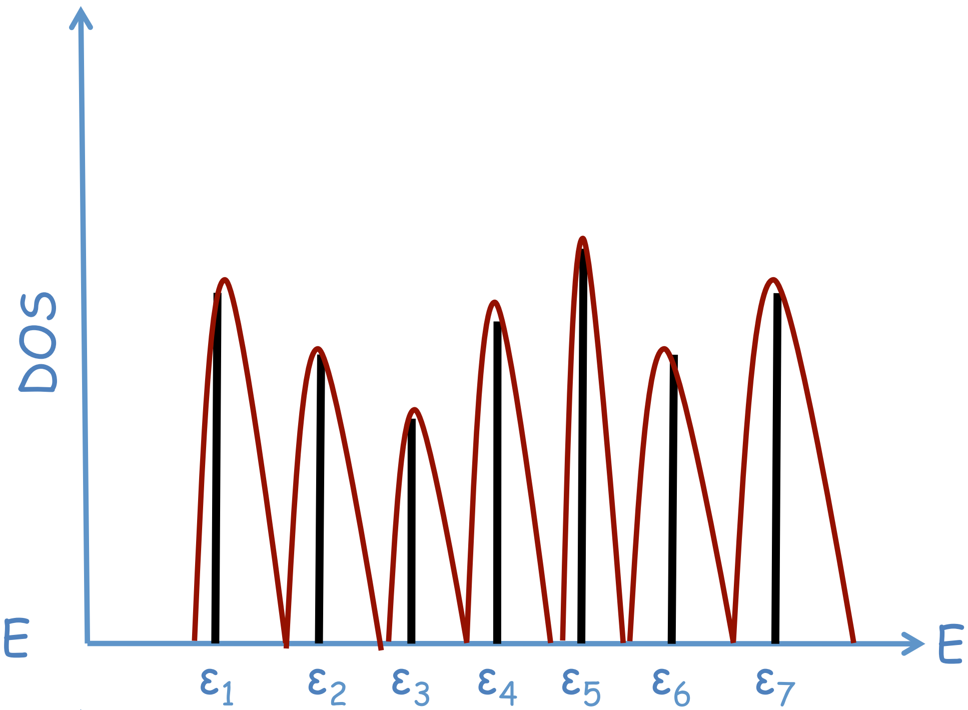
$KTi(SO_4)_2.H_2O$


total
Ti
O
d_{xz}
d_{yz}
d_{xy}
$d_{x^2-y^2}$
$d_{3z^2-r^2}$
t_{2g}
e_g
DOS (states eV^{-1} f.u.$^{-1}$)
Energy (eV)



c
a
b
Fe
Ga1
Ga2
FeGa3
total
Fe
3dxy
3dxz , 3dyz
3d3z²-r²
3dx²-y²
Ga1
4s
4py
4pz
4px
Ga2
4s
4py , 4px
4pz
Density of states (states eV-1 f.u.-1)
Energy (eV)
0=EF
-4
-3
-2
-1
1
2
0
3
6
9
0
1
2
0
0.2
0.4
0
0.2
0.4





$$\text{DOS} = \frac{1}{N} e^{-\left(\frac{E-\varepsilon_i}{\Gamma}\right)^2}$$